

Is Dentistry Safe From AI?

The planning guide — the full scores by track, the six factors behind them, the honest trade-offs, and what to ask before you commit.



2.7

AI EXPOSURE · GENERAL DENTISTRY

where 10 is most at risk

For anyone still choosing — read it straight through.



Before you read this

You're weighing dentistry — or a path into it — and you want to know what that choice actually looks like in an AI economy. That's the whole job of this guide.

It's written to you directly, whether you are the one choosing or a parent thinking it through alongside them. Where it says "you," a parent can read "them"; it's the same decision from either side.

What this is. A facts document, not a recommendation. It sets out what's known, where the evidence is contested, and where we might be wrong — and then how to choose well given all of it.

If you're the parent, a note on how to use it. Don't lead with the scores. Open a conversation with a risk number and one of two things happens: they shut down, or they get defensive about a choice they're already leaning toward. Read it yourself first, then share it as information you both react to, not a verdict you've reached. They're choosing, not you — the job is to widen what's been considered, not to narrow it.

And one thing specific to dentistry. This is the profession where our whole framework has already been tested by history — and it passed. A tier of dentistry was automated more than a decade ago, before anyone was discussing AI, and the profession is fine. The section on the experiment dentistry already ran explains what happened and what it predicts.

The honest issues here are financial, and they are substantial. The section on the financial picture is the one to read carefully.

The short answer

Yes, and dentistry is among the best-protected work we score.

General dentistry scores **2.7** out of 10. Oral surgery scores **2.3**. Dental hygiene scores **3.0**.

Physical work in a patient's mouth, comprehensive licensure, and a relationship built over years — that combination is close to the strongest in this index.

The reasons to think hard about dentistry are entirely financial: **average dental school debt of \$293,900**, and an income picture that depends heavily on whether someone eventually owns a practice. That financial fork — not AI — is most of the decision in front of you.

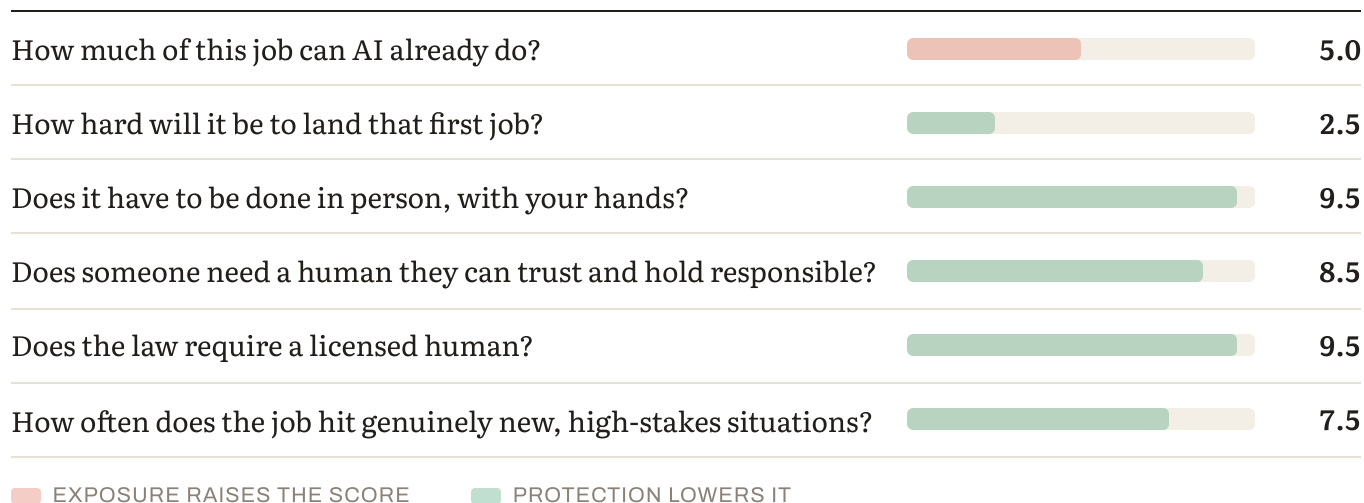
The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Oral & maxillofacial surgery	2.0	2.2	2.3	+0.3	Very low
General dentistry	2.3	2.5	2.7	+0.4	Very low
Dental hygiene	2.6	2.8	3.0	+0.4	Low
Orthodontics	3.0	3.4	3.8	+0.8	Low
Dental laboratory technology	4.8	5.2	5.5	+0.7	Moderate

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, surgery scores 2.1, physical therapy 2.5, bedside nursing 2.8, and entry-level software development 8.1.

Read the table this way, and note the last row particularly — because dental laboratory work is the most instructive number in this guide, and the section on the experiment dentistry already ran explains why.

Why dentistry scores so low — the six factors



Here's how general dentistry answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

Imaging support, scheduling, records, and treatment planning assistance. Not the dentistry.

Radiograph analysis and caries detection. Treatment plan drafting. Clinical notes and records. Appointment scheduling and recall. Insurance claims and coding. Restorative design and milling — which the experiment section covers, because that one already happened.

What resists: working inside a mouth with hand instruments, managing a frightened patient, judgment about what a specific tooth needs, and the relationship that brings someone back for twenty years.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Strong protection. Licensure requires supervised clinical training and school places are capped, so entry is competitive but employment afterwards is secure.

The thing to steer here isn't whether you get hired — it's which tier you land in. Entry is genuinely protected, but the pattern that runs through this whole guide is that the further the work moves from the patient's mouth, the higher the exposure climbs. The way through is to aim at the chairside clinical seat that scores 2.7, with real procedure breadth and a path toward ownership — not to drift toward the administrative and lab-adjacent work that scores 5.5 and above just because it's the rung that happens to be open. That's a set of concrete choices, and the second half of this guide is about making them.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Almost entirely — and with unusual precision. Dentistry is fine motor work in a confined, wet, moving space, on a patient who may be anxious or in pain. This is among the harder physical problems in medicine, not among the easier ones.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and unusually so. Dentistry involves fear more than most healthcare, and trust is what gets people through the door. A licensed dentist is professionally answerable for the outcome.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Comprehensively. Practice, prescribing and procedures are all legally reserved.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Regularly. Every mouth is different, and the judgment about restoration, extraction or referral is made with incomplete information about what's under the surface.

Dentistry already ran this experiment

This is the most useful section in the guide, and it applies well beyond dentistry.

Dental laboratory technology was automated more than a decade before the AI conversation started.

For most of the twentieth century, crowns, bridges and dentures were made by hand by skilled technicians — a genuine craft, learned over years, requiring real artistry. Then CAD/CAM arrived: digital scanning, computer design, milling machines. Much of that work moved from a bench to a screen and a machine, and the number of people needed to do it fell substantially.

Notice what happened next, because it is exactly what our framework predicts.

The profession was fine. Dentists still practice. Patients still get crowns. Demand didn't fall.

But a whole tier of skilled, well-regarded craft work contracted permanently, and the people in it did not simply move up — many moved out. Dental lab technicians now report median pay around \$58,307, and the work that remains is increasingly digital design rather than manual fabrication.

And that is why lab technology still scores 5.5 today while chairside dentistry scores 2.7. The exposure isn't in front of it. Much of it has already happened.

The general lesson, and it is the single most transferable idea in this index: automation does not usually destroy a profession. It removes a tier of it — and which tier you are in matters more than which profession you choose.

Three practical consequences worth carrying to any career:

The tier that goes first is the one with **structured inputs and a physical output that a machine can produce** — a crown from a scan, a document from a template, an image from a brief.

The tier that survives is the one **with the patient, the client, or the accountability** in front of it.

And the displacement is usually invisible from outside the profession. Nobody outside dentistry noticed the lab tier contract. The same is now happening inside law, accounting and design, and it looks the same from the outside: the profession appears healthy, and one layer of it has quietly stopped being a career.

What the trend shows

General dentistry: 2.3 → 2.5 → 2.7. Up 0.4 in three years, and all of it administrative.

Orthodontics: 3.0 → 3.4 → 3.8. Up 0.8 — the fastest-moving track in the profession, driven by treatment planning, aligner design and remote monitoring being genuinely automatable.

Dental lab: 4.8 → 5.2 → 5.5. Continuing a displacement that started years ago.

The gap widened from 2.8 points to 3.2. The pattern holds exactly: the further the work moves from the patient's mouth, the higher the exposure climbs.

The financial picture — and this is the real decision

Dentistry is one of the most expensive qualifications available, and the numbers deserve careful reading.

The debt:

- Average dental school debt is **\$293,900**, per the ADA's Survey of Dental School Seniors.
- The spread is enormous: public schools typically **\$200,000–\$260,000**, private around **\$321,000**, and the most expensive programs reach **\$625,000–\$700,000**.
- One institution's program cost alone is reported at **\$492,820**.

The income, and the gap inside it:

- BLS reports a mean annual wage of **\$166,810** for general dentists.
- ADA Health Policy Institute puts median net income for practice owners closer to **\$239,000**.
- ADA HPI analysis of Department of Education data puts average net income for general practitioners **five years after graduation at \$171,062**.
- New dentists typically earn **\$150,000–\$200,000** in their first year.
- Specialists out-earn generalists by roughly 30–90%.

That \$166,810-versus-\$239,000 gap is the whole story. Dentistry's income model assumes eventual practice ownership, and a student should understand that at seventeen rather than at thirty-two.

And one statistic that says something the others don't. ADA data indicates that around **35% of dental school costs are financed without education loans**, with a further 17% covered by family. Dentistry draws disproportionately from families able to fund it, which means the "average debt" figure understates the burden for students who borrow the whole amount.

The practical instruction is the same as for physical therapy: the spread between the cheapest and most expensive accredited program is larger than most people's mortgages. **Attend the least expensive accredited school you get into.**

The alternative that deserves proper attention

Dental hygiene, and the numbers are genuinely striking.

- Median pay of **\$98,100** (BLS OEWS, May 2025) — slightly above the registered nurse median of \$97,550
- **7% projected growth** through 2034, with about **15,300 openings a year**
- Around **222,740** practicing hygienists, 94% of them in dentists' offices
- **No night or weekend shifts**, unlike nursing
- Program debt typically **\$30,000–\$80,000** — an order of magnitude below dental school
- And it scores **3.0** on our index, against 2.7 for dentistry

Two honest caveats, because this isn't a free lunch:

Part-time work is the norm rather than the exception. In ADA Health Policy Institute's survey of 5,122 hygienists, 28.8% worked under 32 hours a week against 53.7% full-time — roughly one in three part-time. Annual salary figures assume full-time, year-round hours that many hygienists don't work.

And benefits follow from that. BLS notes that vacation, sick leave and retirement contributions may only be available to full-time staff, and ADA research finds health insurance and paid leave are far from universal in private dental practices.

There is also less of a ladder. Nursing offers a longer advancement path; hygiene is relatively flat.

But for someone drawn to clinical dental work rather than to diagnosis, business ownership and treatment planning — two to three years and \$50,000 of debt against seven-plus years and \$293,900 is a comparison worth making properly.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Very strong	Most fragile long-term	The protection physical AI targets — though a wet, confined, moving mouth is among the harder robotics problems.
Regulatory / licensing	Strong	Most durable	Practice and procedures are legally reserved.
Human trust / accountability	Strong	Durable	Dentistry involves fear; trust is what brings people back.
Judgment under novelty	Good	Eroding slowly	Diagnostic imaging support is advancing; chairside judgment is not.

The honest read. Dentistry has all four protections operating together, which is rare — and the historical evidence in the experiment section shows the framework working prospectively. The automation arrived where the model says it should: at the structured, machine-producible tier.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Reading the radiograph unaided	First-pass caries detection arrives; the dentist adjudicates
Writing clinical notes	Reviewing an ambient draft
Fabricating restorations by hand	Already gone — CAD/CAM did this a decade ago
Drafting the treatment plan	Reviewing a generated plan and judging whether it fits
Insurance coding and claims	Substantially automated
Working inside the mouth	Unchanged
Managing a frightened patient	Unchanged

Human strengths that will matter most

1. **Fine motor work in a confined space** — the core of the craft, and among the hardest physical problems in medicine. 2. **Managing fear** — dentistry involves more patient anxiety than most healthcare, and handling it well is a genuine and undersupplied skill. 3. **Judgment on what a specific tooth needs** — restore, extract, refer, or wait. 4. **Long-term relationships** — the practice model depends on people coming back for decades. 5. **Verification** — knowing when an automated caries detection or treatment plan is wrong.

Notably absent: the hand fabrication of restorations — the lab-bench craft that CAD/CAM already took. That was once a respected marker of skill, and it is the clearest thing on the "already gone" list.

And underneath all five, two traits the score can't grade: curiosity and adaptability. The specific tools — imaging AI, chairside CAD/CAM, digital treatment-planning software — will turn over many times across a career. The dentists who compound are the ones who stay genuinely curious about the work and re-tool fastest, rather than defending the one way of working they trained on. In a profession where the digital workflow keeps moving into the operatory, that flexibility is itself a form of protection — and increasingly it's what the strongest practices are built around.

Other things to consider about this job

What's genuinely good about it

Satisfaction is high and unusually consistent. Survey data puts career satisfaction around 74%, with 85% saying they would choose their specialty again — figures at the higher end of healthcare.

And the working pattern is better than most of medicine. Dentists report averaging around 36 hours a week, with no routine night or weekend work in most general practice — a genuine difference from medicine and nursing.

What dentists report as rewarding:

- **Immediate, visible results** — you fix something and see it fixed
- **Combining manual craft with clinical judgment**, which few careers offer
- **Autonomy**, with practice ownership a normal destination rather than an exotic one
- **Long relationships** with patients and families across decades
- **Genuine control over working pattern**, particularly for owners

What's genuinely hard about it

The debt, covered in the financial section, and it is the dominant issue.

The physical toll is real and under-discussed. Dentistry is performed in sustained awkward postures, and musculoskeletal problems — neck, back, shoulders, wrists — are a recognized occupational hazard that can shorten careers.

Patient anxiety is a constant. A meaningful share of patients arrive frightened, and managing that repeatedly is emotionally demanding in a way outsiders rarely appreciate.

The business is a second job. Ownership is where the income is, and running a practice means staff, overheads, insurance negotiation and marketing — none of which dental school teaches well.

And insurance economics are a persistent frustration, with reimbursement pressure a leading complaint in the profession.

Who this work suits — and who it doesn't

Dentistry tends to suit people who:

- **Have genuine manual dexterity** and enjoy precise physical work
- **Are comfortable with people who are frightened** — this is most of the interpersonal job
- **Want immediate, visible results** rather than long-horizon outcomes
- **Are interested in running a business**, or at least willing to be
- **Can work in confined, repetitive physical conditions** for decades

It's harder for people who:

- Want **variety of setting** — it's largely the same room, the same chair
- Are uncomfortable **causing discomfort**, which is unavoidable
- Have no interest in the **commercial side**, given where the income sits
- Need **intellectual novelty** daily — much of general practice is repetitive by design

What else is moving this market

Consolidation is reshaping the profession. Dental support organisations have grown substantially, and the proportion of dentists in traditional solo private practice has fallen. That changes the ownership assumption underpinning the income model, and it's the most important structural shift in the profession right now.

Demand is stable and demographically supported, with an ageing population retaining more natural teeth than previous generations.

And the specialty premium is significant — specialists out-earn generalists by 30–90%, at the cost of additional years and additional debt.

The AI-native advantage — what to actually do about it

The situation: dental AI is arriving as imaging support, treatment planning assistance and administrative relief. The clinical work is untouched. A good program should be building this fluency deliberately, not banning the tools or ignoring them.

WHAT AN AI-NATIVE DENTIST LOOKS LIKE

1. Verify. Knowing when an automated caries detection is wrong in either direction. A false positive means unnecessary treatment on a healthy tooth, which is a clinical and ethical problem, not just a technical one.

2. Use imaging support properly. These systems are genuinely good and genuinely confident when wrong.

3. Run the digital workflow. Intraoral scanning, CAD/CAM chairside milling, digital treatment planning. The lab tier moved into the practice — a dentist who can operate that workflow captures value that used to leave the building.

4. Handle patient AI use. Patients arrive having consulted symptom checkers and comparison sites, often about cost. Working with that rather than dismissing it is a new communication skill.

5. Take the business seriously. Everything in the financial section points at ownership, and practice management is the least-taught and most consequential skill in the profession.

CONCRETE PREPARATION

Before applying: compare total program cost seriously. The spread between the cheapest and most expensive accredited school exceeds \$400,000, for the same license.

During: take practice management and business content seriously rather than treating it as peripheral. Get exposure to digital workflows, and to a range of practice settings including DSO-affiliated ones.

Entering: understand associate versus ownership economics before signing anything, and get any employment contract reviewed.

The routes in

ROUTE	TYPICAL LENGTH	NOTES
DDS / DMD	4 years post-degree	The standard route. Cost varies by more than \$400,000 between schools.
Dental hygiene	2–3 years	Median \$98,100, debt \$30K–\$80K, no nights or weekends. Genuinely under-considered.
Dental assisting	Under a year to 2 years	Fastest entry; median around \$36,503.
Dental therapy	Varies, limited jurisdictions	An expanding mid-level route in some regions.
Specialty training	+2–6 years post-DDS	30–90% income premium at the cost of more years and more debt.

Where a dental qualification can lead later

Clinical practice across general and specialist areas, practice ownership and multi-site groups, academia and teaching, public and community dental health, industry and device companies, and dental research.

The pattern in this index holds: the further from the chair, the higher the exposure. Laboratory and administrative roles score 5.5 and above against chairside practice at 2.7. That gradient is the compass for every choice below.

What to look for in a dental program

1. **Total cost** — and calculate the debt-to-income ratio yourself. This matters more than anything else on this list.
2. **Accreditation**, which is non-negotiable.
3. **Clinical volume** — how many procedures does a student actually perform? This varies enormously between schools and it determines competence at graduation.

4. Digital workflow and practice management content. Both are where the profession is going and both are commonly underweighted.

5. Board pass rates and first-destination outcomes.

Questions to ask a dental program

On cost — ask these first:

- What is the total cost of attendance for the full program, including living expenses?
- What is the median debt at graduation for your graduates?

On clinical training:

- How many procedures does a typical student complete before graduating?
- What range of settings do students see, including community and DSO practices?

On the curriculum:

- How much practice management and business content is compulsory?
- How does the program teach digital workflow — scanning, CAD/CAM, treatment planning?
- Where do students learn to verify AI-assisted imaging analysis?

Red flags: evasiveness about total cost or graduate debt; low clinical procedure volume; practice management treated as optional; no digital workflow training.

The career-hunting checklist

Choosing well is step one. This is what turns a good choice into a good outcome once you're in — worth reading now, because the best time to pick a program is knowing exactly what you'll need to do with it. The first list is for the degree and training years; the second is for the job hunt that follows.

During the degree / training — build what can't be automated

- **Treat total cost and the ownership path as the core decision.** The income model assumes eventual ownership — \$166,810 employed against roughly \$239,000 for owners. Attend the least expensive accredited school you get into, because the spread between programs exceeds \$400,000 for the identical license.
- **Maximise clinical volume.** The protected core (2.7) is the hands-in-the-mouth work. The more procedures you actually perform before graduating, the more of that core you own — and it varies enormously between programs, so chase it.
- **Learn the digital workflow while it's free.** Intraoral scanning, chairside CAD/CAM milling, digital treatment planning. The lab tier (5.5) moved into the practice; the dentist who runs that workflow captures value that used to leave the building.
- **Take practice management seriously, not as the dull option.** It is the least-taught and most consequential skill in the profession, and dental school teaches it poorly — so pursue it deliberately.
- **Build verification into how you read imaging.** Automated caries detection is confident when wrong; knowing when to overrule it — a false positive means treating a healthy tooth — is a clinical skill, not a technical one.
- **Feed curiosity and adaptability on purpose.** The tools — imaging AI, CAD/CAM, treatment-planning software — will turn over repeatedly; the dentist who stays curious and re-tools fastest compounds.
- **Keep the frame wide.** Properly vet the alternatives — dental hygiene, dental therapy, the specialty routes — before committing. Dental hygiene in particular is chronically under-considered: \$98,100 median, \$30K–\$80K debt, a 3.0 score.

During the job hunt — aim at the chair, don't just apply

- **Target the protected lane on purpose.** A chairside clinical seat with real procedure breadth (2.7), not a role weighted toward the administrative or lab-adjacent tier that scores 5.5 and above.
- **Read the setting, not just the offer.** The gradient is fixed: the further from the patient's mouth, the higher the exposure. A seat built around routine, machine-producible or purely administrative throughput points the wrong way.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. A practice or group still building its model around the machine-producible or administrative tier may simply be behind the curve — and that's exactly the rung that contracted in the lab experiment. Availability can be a lagging indicator, not a green light.

- **Understand associate versus ownership economics before signing.** Ownership is where the income is; get any employment contract reviewed, and know what path to ownership the role actually offers.
- **Weigh the specialty question with eyes open.** Orthodontics is the fastest-moving track (3.8), driven by treatment planning, aligner design and remote monitoring being automatable — a 30–90% premium bought with more years and more debt.
- **Weigh trajectory over the first-year number.** A seat that builds clinical breadth, digital-workflow fluency and a path to ownership beats a higher first-year figure in a lane that's drifting away from the chair.

Go deeper — further reading

- **[ADA Health Policy Institute — dental practice research](#)** — American Dental Association. The profession's own data on income, practice ownership, consolidation and workforce. The primary source for most of the financial section, and free.
- **[Dentists — Occupational Outlook Handbook](#)** — US Bureau of Labor Statistics. Median pay, projected growth and entry requirements. Read the **Dental Hygienists** page alongside it — that comparison is the most useful twenty minutes available on this decision.
- **[Average dental school debt](#)** — program-by-program cost data, including the finding that around 35% of dental school costs are financed without loans. Worth reading before choosing where to apply.

The counterargument — read this one:

The strongest case against our 2.7 isn't that AI will replace dentists. It's that dentistry already showed us what happens, and we may be under-reading it.

On this view, the lab tier's disappearance is not reassurance — it is the warning. A skilled, respected, decades-old craft inside dentistry was substantially automated within about ten years of the enabling technology maturing, and the people in it did not move up. If milling machines could do that to lab work, the argument runs, chairside robotics could eventually do something similar to routine restorative procedures, and our 2.7 reflects present capability rather than trajectory.

Where we land: we take it seriously, and it's why physical protection is the factor we rank least durable everywhere in this index. What holds our score is the specific difficulty of

the environment — a wet, confined, moving mouth belonging to an anxious person is a materially harder robotics problem than a milling machine working from a scan, and the regulatory and trust protections would remain even if the manual problem were solved.

But the honest position is that dentistry demonstrates our framework works — including when the answer is uncomfortable.

Bottom line

On AI exposure, dentistry is among the safest work we measure. Chairside practice at 2.7 is protected by all four of our factors at once — fine motor work in a mouth, comprehensive licensure, deep patient trust, and clinical judgment on incomplete information.

And dentistry gives this whole index its clearest proof. The lab tier was automated more than a decade before anyone discussed AI, the profession was fine, and a tier of skilled craft work contracted permanently. Automation removes a tier, not a profession — and which tier you're in matters more than which profession you choose.

The real decision is financial. Average debt of **\$293,900**, with a spread between schools exceeding \$400,000 for the identical license — against a mean wage of \$166,810 that rises to around \$239,000 for practice owners. The income model assumes ownership, and that assumption is worth understanding at seventeen rather than at thirty-two.

So the useful advice is: attend the least expensive accredited school you get into, because the spread dwarfs everything else. Take practice management seriously, since ownership is where the income sits. Learn the digital workflow, because the lab tier moved into the practice and whoever runs it captures the value.

And look properly at **dental hygiene** before committing — median pay slightly above a registered nurse, no nights or weekends, debt an order of magnitude lower, and a score of 3.0. With the honest caveat that a third of hygienists work part-time and benefits are patchy — but for someone drawn to clinical dental work rather than to business ownership, that comparison deserves an hour rather than a dismissal.

Questions worth sitting with

Whether you're the one choosing or a parent thinking it through together, these are the conversations that lead to a better decision. Read "you" as whoever the choice is about.

About the work itself

1. What made dentistry appealing? Push past "I want to work in healthcare" to something specific. 2. **Do you actually enjoy precise manual work?** This is the core of the job and it's testable. 3. Are you comfortable with people who are frightened, repeatedly? 4. Do you realize most general practice is the same room, the same chair, and largely repetitive by design?

Reacting to the findings

5. Chairside dentistry scores 2.7 and lab technology 5.5. Does the difference make sense to you? 6. **The lab tier was automated a decade before the AI conversation.** What does that suggest about other careers on the table? 7. Orthodontics is the fastest-moving track in the profession. Does that affect any specialty thinking?

The money conversation, and this is the important one

8. **Have you compared total cost across the schools you're considering?** Average debt is \$293,900 and the spread between programs exceeds \$400,000. 9. Do you understand that the income model assumes eventual practice ownership — \$166,810 employed against roughly \$239,000 for owners? 10. Are you interested in running a business? Because that's where the money is, and dental school teaches it poorly. 11. **Have you looked at dental hygiene properly?** \$98,100 median, \$30K–\$80K debt, no nights or weekends — with the part-time and benefits caveats understood.

Testing it against reality

12. **The most valuable single action:** shadow in two very different practices — a private owner-operated one and a DSO-affiliated one. The economics and the culture are genuinely different. 13. Find a dentist five years qualified and ask directly about their loans and whether they'd choose the same school.

Widening the frame

14. If dentistry weren't available, what would you choose? The answer usually reveals what you're drawn to. 15. Have you considered **dental hygiene or dental therapy** seriously, rather

than as a fallback? And what about **medicine or physical therapy** — both score similarly well with different financial profiles.

This is analysis and informed judgment about a fast-moving situation. For dentistry specifically, we've flagged that the profession's most serious risk is financial rather than technological, and that the automation of dental laboratory work more than a decade ago is the clearest historical test of the framework used throughout this index.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Dentistry

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

ORAL & MAXILLOFACIAL SURGERY — 2.3 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.2	4.6	4.8	1.68
Entry-path erosion	15%	1.2	1.5	1.8	0.27
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	2.3 → 2.3

GENERAL DENTISTRY — 2.7 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.7	5.1	5.6	1.96
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.3	9.3	9.3	0.1
Human trust / accountability	15%	9.5	9.5	9.5	0.07
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	2.69 → 2.7

DENTAL HYGIENE — 3.0 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.7	5.2	5.7	1.99
Entry-path erosion	15%	1.8	2.0	2.2	0.33
Physical / embodied	15%	9.2	9.2	9.2	0.12
Human trust / accountability	15%	9.0	9.0	9.0	0.15
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	3.0 → 3.0

ORTHODONTICS — 3.8 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.6	6.7	7.7	2.69
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	8.5	8.5	8.5	0.22
Human trust / accountability	15%	9.0	9.0	9.0	0.15
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	3.79 → 3.8

DENTAL LABORATORY TECHNOLOGY — 5.5 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.5	4.4	5.1	1.78
Entry-path erosion	15%	4.5	5.0	5.5	0.82
Physical / embodied	15%	5.0	5.0	5.0	0.75
Human trust / accountability	15%	4.0	4.0	4.0	0.9
Regulatory / licensing	10%	3.0	3.0	3.0	0.7
Judgment under novelty	10%	4.5	4.5	4.5	0.55
				Total	5.51 → 5.5

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Oral & maxillofacial surgery

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	4.2	1.2	0.35	2.0
2025	4.6	1.5	0.35	2.2
Fall 2026	4.8	1.8	0.35	2.3

Movement decomposition: automatability contributed **+0.21** and entry-path erosion **+0.09**, for a total of **+0.3** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.